

and drooping ears and prominent dewlap. It belongs to the family Bovidae in the order Artiodactyla.

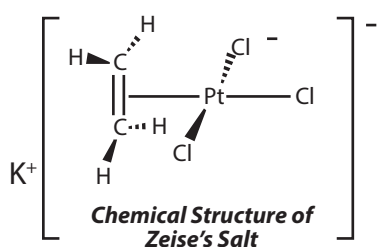
ZEEMAN DISPLACEMENT: The separation in wave numbers of adjacent spectral lines in the normal Zeeman effect in a unit magnetic field. It is expressed in centimetre-gram-second Gaussian units and the formula is $\frac{e}{4\pi m c^2}$, where e and m are the charge and mass of the electron. It is approximately $4.67 \times 10^{-5} (\text{centimetre})^{-1}(\text{gauss})^{-1}$.

ZEEMAN EFFECT: The splitting of spectral lines when the source of the spectrum is placed in a magnetic field, discovered by P. Zeeman in 1896. Each kind of spectral term has its characteristic mode of splitting and the types of terms are most definitely identified by this property. In the normal Zeeman Effect, a single line is split into three if the field is perpendicular to the light path or two lines if the field is parallel to the light path. The anomalous Zeeman Effect is a complicated splitting of the lines into several closely spaced lines. This effect is explained by quantum mechanics in terms of electron spin. The effect allows an evaluation of the ratio of charge to mass of the electron and an evaluation of its precise magnetic moment.

ZEILINGER'S PRINCIPLE: The principle which states that any elementary system carries just one bit of information. This was extended to develop several aspects of quantum mechanics.

ZEIN: A low-molecular-weight simple protein found as a major storage protein in maize grains.

ZEISE'S SALT [POTASSIUM TRICHLORO(ETHENE) PLATINATE(II)]: A chemical compound with the formula $K[PtCl_3(C_2H_4)] \cdot H_2O$, which is a complex of platinum and ethene in which the platinum coordinates to the pi bond of the ethene. The compound is commercially available as a hydrate, prepared from potassium tetrachloroplatinate(II) ($K_2[PtCl_4]$) and ethylene in the presence of a catalytic amount of $SnCl_2$.



ZEISEL REACTION: A technique of determining the number of methoxy ($-OCH_3$) groups in an organic compound, developed by the Czech chemist, Simon Zeisel (1854-1933) in 1886 and named after him. The compound is heated with excess hydriodic acid, forming an alcohol and iodomethane: $R-O-CH_3 + HI \rightarrow ROH + CH_3I$. The iodomethane is distilled off and led into an alcoholic solution of silver nitrate, where it precipitates silver iodide. This is filtered and weighed and the number of iodine atoms and hence methoxy groups can be calculated.

ZENER DIODE: A type of semiconductor diode, named after C. Zener, who first proposed electronic tunnelling as a mechanism of electrical breakdown in insulators. It is a special junction-type diode, using silicon, in which the voltage across the junction is independent of the current through the junction. It is made up of a p-n junction of two different kinds of semiconductor material with high doping concentrations on either side of the junction with a very sharp voltage

breakdown as reverse bias is applied. When the device is conducting, the diode acts as a rectifier until the applied reverse voltage reaches a certain value known as the Zener breakdown voltage. This is as a result of electrons being excited directly from the valence band into the conduction band. Zener diodes are used as voltage regulators.

ZENITH: 1. The uppermost point of the celestial horizon immediately above the observer or the point of the celestial sphere that is directly over the observer and 90 degrees from all points on that person's horizon. If an observer is at a pole, then the observer's zenith is a celestial pole; but for observers at mid-latitudes the zenith is a point in the sky that corresponds to a changing right ascension but a constant declination as the sky rotates overhead. The zenith distance of an object in the sky is the angle across the sky from the zenith to the object. The zenith distance is 90° minus the altitude of an object above the horizon. The opposite is **nadir**, which is the point on the celestial sphere directly below the observer and opposite the zenith. 2. The highest point reached by an astronomical object.

ZEOLITE: General term for a family of minerals called tectosilicates, which is a natural or synthetic hydrated aluminosilicate with an open three-dimensional crystal structure, in which water molecules are held in cavities in the lattice. Zeolite is composed of sodium, calcium and potassium, usually found in veins and cavities of basic igneous rocks, especially basalt. They are characterised by ready loss or gain of water. Examples of zeolites are stilbite, chabazite, natrolite and analcime. Zeolites are used for separating mixtures by selective absorption and as water-softening agents in an ion-exchange method called zeolite process.

ZEOLITE CATALYST: Hydrated aluminium and calcium or sodium silicates made with controlled porosity. It is used as a catalytic cracking catalyst in petroleum refineries. It may also be loaded with catalyst for other chemical reactions.

ZEPTO-: A prefix with the symbol z used in the metric system to indicate 10^{-21} . For example, 1 zeptosecond (zs) = 10^{-21} second.

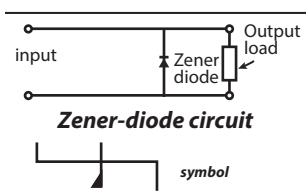
ZEREWITINOFF REAGENT: A light-coloured methyl magnesium iodide-*n*-butyl ether solution that reacts rapidly with moisture and oxygen. It is used to determine water, alcohols and amines in inert solvents.

ZERMELO-FRANKEL THEORY (ZF): The theory, named after mathematicians Ernst Friedrich Ferdinand Zermelo (1871-1953), German logician and mathematician and Abraham Fraenkel (1891-1965), German-born Israeli mathematician. It is a one-sorted theory in first-order logic. The signature has equality and a single primitive binary relation, set membership, which is usually denoted $\in$. The formula $a \in b$ means that the set a is a member of the set b , also read as " a is an element of b " or " a is in b ". The theory with the addition of the axiom of choice is written ZFC, the C being choice.

ZERMELO'S AXIOM (AXIOM OF CHOICE): The axiom of set theory stating that from every family of disjoint sets, a set can be constructed containing exactly one element of each of the given family of sets.

ZERMELO'S THEOREM: In logic, also called well-ordering theorem, it is the non-constructive set theoretic result, equivalent to the axiom of choice or Zorn's lemma, that for any set there is a binary relation under which it is well-ordered.

ZERO: 1. The figure nought (0) or the point between the positive (+) and the negative (-) on a scale; or the numerical symbol 0, representing the absence of any quantity or magnitude. When added to or subtracted from any number, it results in that number retaining its



value without a change, it is therefore known as the additive element. It is also the starting or centre point for values on a counter, scale or gauge; or the temperature indicated by 0 on a thermometer scale or the lowest possible point or degree. Any number multiplied by zero gives zero. Zero multiplied by or divided by any number (other than zero) is still zero. The expression zero divided by zero (0/0) has no defined value and is called an **indeterminate form**. The introduction of zero into the decimal system was the most significant achievement in the development of a number system in which calculation with large numbers was feasible. Without it, modern astronomy, physics and chemistry would have been unthinkable as we know them. 2. The cardinality of the empty set. 3. The identity element for any additive operation, so that its sum with any other element is that other element, such as the matrix whose elements are all 0. 4. An element of a ring, of which the product with any other is that zero element. It is often denoted 0. 5. Zero of a function is a value of the argument of a function at which the value of the function is zero. For example, $x^2 + 2x$ has a zero at $x = -2$. 6. The least element of a lattice or partially ordered set, such as the empty set in the poset of subsets of a given set. Its symbol is Λ .

ZERO DIVISORS: Non-zero elements in a ring of which the product is zero such as the matrices

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad A \text{ and } B \text{ are zero}$$

divisors within the ring of 2×2 matrices defined with the usual matrix addition and multiplication. A ring is a left zero divisor if there exists a nonzero B such that $AB = 0$. Similarly, a nonzero element a of a ring is a right zero divisor if there exists a nonzero c such that $ca = 0$. An element that is both a left and a right zero divisor is simply called a zero divisor. If multiplication in the ring is commutative, then the left and right zero divisors are the same. A nonzero element of a ring that is neither a left nor right zero divisor is called regular.

ZERO ELEMENT: One of several generalisations of the number zero to other algebraic structures. For example, the element 0 is a zero element of a group, if $a + 0 = a$ and $0 + a = a$ for all elements a .

ZERO ERROR: The error that arises, when the scale reading of a measuring instrument has an initial pointer reading, which is significantly different from its calibrated zero mark (reading). Examples of instruments that can give zero errors include the following: thermometer, micrometer screw gauge, vernier caliper, ammeter and voltmeter. Averaging the results of several measurements cannot reduce zero error.

ZERO EXPONENT PROPERTY: Any non-zero number raised to an exponent equalling zero equals one, for example, $5^0 = 1$.

ZERO FIELD SPLITTING: The separation of multiplet sub-levels in the absence of an external magnetic field.

ZERO GRAZING: A system of feeding cattle or other livestock in which freshly cut forage is brought daily to animals that are permanently housed instead of being allowed to graze.

ZERO HOUR (H HOUR): The scheduled time at which operations are to begin, especially a combat operation of great size.

ZERO INSERTION FORCE SOCKET (ZIF SOCKET): In computer science, it is a socket on a computer's motherboard designed by Intel that enables a computer processor to be easily removed or upgraded by the use of a small lever.

ZERO MATRIX: The $m \times n$ matrix with all of its entries zero. Examples

$$\text{are: } \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

ZERO MEASURE: 1. Also known as **null measure**, it is a measure of a subset of n -dimensional Euclidean space for which, for every $\epsilon > 0$, there is a covering of the subset by rectangles whose volume is less than ϵ . Any set of 0 measure is called a null set. 2. A measure, μ , such that $\mu(E) = 0$ for every measurable set, E .

ZERO OF A FUNCTION: A value of the argument of a function at which the value of the function is zero. For example, the function $f(x) = \frac{x+2}{2x+3}$

has a zero of $x = -2$, i.e., when $f(-2) = \frac{-2+2}{2(-2)+3} = \frac{0}{-1} = 0$. Therefore,

a function $f(x)$ has a zero, when $f(x) = 0$.

ZERO PLACEHOLDER: A term used in the place value system. For example, 0 is the zero placeholder in the number 8209.

ZERO POINT: Value of the pH of a solution, which in combination with a stated outer reference electrode, gives zero emf from the operational cell.

ZERO POINT OF SCALE: The rest point of the properly adjusted balance with no load on the pans and the rider (or chain) in the zero position.

ZERO PROPERTY OF DIVISION: The quotient of zero divided by any non-zero number equals zero.

ZERO PROPERTY OF MULTIPLICATION: The product of any real number and zero equals zero.

ZERO RING (TRIVIAL RING): A ring, denoted $\{0\}$ or 0 consisting of only a single element with the ring operations multiplication ($\cdot$) and addition ($+$) defined so that $0 + 0 = 0$ and $0 \cdot 0 = 0$. It is a commutative ring with an identity.

ZERO SET: The set of values for which a given non-zero analytic function is zero. In other words, the zero set of the function f is the subset of a set X on which $f(x) = 0$. This set is at most countable. A **co-zero** set in a set X is a subset whose complement is a zero set.

ZERO SLOPE: The slope of horizontal line where the amount of vertical change equals zero and the amount of horizontal change equals a non-zero real number, yielding a slope ratio whose numerator equals zero.

ZERO TILLAGE: A cultivation technique in which the soil is disturbed only along the slit or hole into which seeds are planted and herbicides are used to control weed growth. The advantages include: (i) decreased rate of soil erosion; (ii) reduced need for equipment, fuel and fertiliser; (iii) significantly less time required for tending crops. The method also improves soil-aggregate formation, microbial activity in the soil and water infiltration and storage.

ZERO VECTOR (TRIVIAL VECTOR): A vector with neither magnitude nor direction; its length is zero and therefore does not point in any particular direction. An example is the two-dimensional

$$\text{vector, } \vec{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

ZERO-ONE LAW: In probability theory, it is the law which states that a tail event will either almost surely happen or almost surely not happen, that is a sequence of independent random variables is either 0 or 1. It is named after Andrey Nikolaevich Kolmogorov (1903-1987), Soviet mathematician.

ZERO-ORDER REACTION: A reaction for which the reaction rate is independent of the concentration of the reactants. The rate is expressed as $\text{rate} = k$, with k having the units of M/sec. An example is a photochemical reaction in which the rate is determined by the intensity of light.

ZERO-POINT ENERGY: The energy remaining in a substance at the absolute zero of temperature (0 K). This agrees with the quantum theory in which a particle oscillating with simple harmonic motion

does not have a stationary state of zero kinetic energy. Moreover, the uncertainty principle does not allow such a particle to be at rest at exactly the centre point of its oscillations.

ZERO-SUM GAME: A game in which the total pay-off is zero and the payoff for one player is the negative of the pay off for the other. In other words, the gain of one player is equivalent to the loss of the other.

ZERO-ZERO (0-0) ABSORPTION/EMISSION: A purely electronic transition occurring between the lowest vibrational levels of two electronic states.

ZEROISE (ZERO; ZEROIZE): To set to zero, especially to initialise the values of the variables in an algorithm or computation to zero.

ZEROTH: The zeroth of a sequence of terms is the term that is numbered zero and precedes the first term that is numbered 1. For example, in the sequence $a_0, a_1, a_2, \dots, a_n$ is the zeroth.

ZEROTH LAW OF THERMODYNAMICS: A law which states that if two systems are separately found to be in thermal equilibrium with a third system, the first two systems are in thermal equilibrium with each other, that is, all three systems are at the same temperature.

ZETA FUNCTION (RIEMANN ZETA FUNCTION; EULER-RIEMANN ZETA FUNCTION): In mathematics, it is the special function, denoted $\zeta(s)$ that is defined over the complex plane for one complex variable, s . The sum of the infinite series $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, is infinite when $s = 1$; when s is greater than 1, it converges to a finite

number as successive terms are added; if s is less than 1, the sum is again infinite. It is a very important function in number theory.

ZETTA-: A prefix with the symbol Z , used in the metric system to indicate 10^{21} . For instance, 1 zettametre (Zm) = 10^{21} metres.

ZIDOVUDINE: A drug which inhibits replication of viral DNA, used in the treatment of AIDS and prevention of HIV transmission from an infected mother to her baby. It is not a cure for AIDS but was thought to be effective in suppressing the causative virus (HIV).

ZIEGLER CATALYST (ZIEGLER NATTA CATALYST): A special catalyst developed to produce stereospecific polymers and made by mixing certain transition elements or salts such as titanium(IV) chloride (TiCl_4) with an organometallic compound such as triethylaluminium ($\text{Al}(\text{C}_2\text{H}_5)_3$). It is used in the Ziegler process.

ZIEGLER PROCESS: An industrial process for making high-density polyethylene, using catalysts of titanium(IV) chloride (TiCl_4) and aluminium alkyls, for example, triethylaluminium, $\text{Al}(\text{C}_2\text{H}_5)_3$. The process was introduced in 1953 by the German chemist Karl Ziegler (1898-1973). It allowed the manufacture of polythene at lower temperatures (about 60 °C, 333 K) and pressures (about 1 atm) than used in the original process.

ZIF SOCKET (ZERO INSERTION FORCE SOCKET): In computer science, it is a socket on a computer's motherboard designed by Intel that enables a computer processor to be easily removed or upgraded by the use of a small lever.

ZIG-ZAG PROJECTION: A stereochemical projection for an acyclic molecule (or portion of a molecule), where the main chain is represented by a zig-zag line in the plane and the substituents are shown above or below the plane.

ZIGZAG HARROW: A light harrow used for final seedbed work and also for covering sown seeds. The frames are zigzag in shape, with short tines bolted to them.

ZIMM PLOT: A graphical determination of the root-square-mean end-to-end distances of coil-like polymer molecules during scattered-light photometric analysis.

ZINC: A grey, hard, brittle metallic element, the first element in Group 12 of the Periodic Table with the symbol Zn and atomic number 30. Zinc melts at about 420°C (693 K), boils at about 907 °C (1180 K)

and has a relative density of 7.14. It is extremely brittle at ordinary temperatures, but becomes malleable between 120 °C and 150 °C (393 K and 423 K) and may be rolled into sheets between heated rollers. It is reactive towards acids but does not react with cold water. Its chief ore is sphalerite, zinc sulfide or zinc blende. Zinc is the 24th most abundant element in the Earth's crust and has five stable isotopes. It is made by roasting and smelting of its ore. It is used to make alloys such as brass and some types of battery.

ZINC BLENDE (SPHALERITE): Zinc sulfide (ZnS) in mineral form but almost always contains variable iron. It is the chief ore of zinc. It is usually found in association with galena, pyrite and other sulfides along with calcite, dolomite and fluorite. It crystallises in the isometric system and has a hardness of 4 on the Moh's scale and a relative density of 4. It has a resinous lustre and exists in massive or crystalline form and usually contains iron sulfide, which gives it a dark colour. It also occurs in red, yellow or green.

ZINC CHLORIDE: A white crystalline hygroscopic salt with the chemical formula ZnCl_2 , melting point 283°C (556 K) and boiling point 732 °C (1005 K). It is deliquescent and sublimes easily. It is produced commercially by heating metallic zinc in hydrogen chlorine gas. It is used as a catalyst, a dehydrating agent and as a flux in soldering. It is also used to fire proof timber, in dry battery making, vulcanising and galvanising, in oil refining and as a fungicide and catalyst.

ZINC CHLORIDE CELL: A primary cell in which the electrolyte consists of a mixture of ammonium chloride and zinc chloride made into a paste. The negative electrode is made of zinc, as is the outside shell of the cell and the positive electrode is a carbon rod surrounded by a mixture of carbon and manganese dioxide and carbon powder. They are the least expensive primary batteries and come in several traditional sizes such as AA, AAA and used in household devices such as remote controls, flashlights, clocks or transistor radios.

ZINC FINGER: A structural motif organised around a zinc ion and found in many gene-regulatory proteins. It consists of a finger-like fold of amino acid at the base of which lie two cysteine and histidine residues. They can be classified into several different structural families and typically function as interaction modules that bind DNA, RNA, proteins or small molecules.

ZINC GROUP: A group of elements belonging to the transition metals in Group 12 of the Periodic Table consisting of zinc, cadmium and mercury.

ZINC ORE: A mineral from which zinc is extracted, for example, zinc blende or sphalerite, the chief ore of zinc and zincite.

ZINC OXIDE: A white powder when cold and yellow when hot with the chemical formula ZnO and melting point 1975 °C (2248 K). It occurs in nature as the mineral zincite. It can also be made by oxidising hot zinc in air. It is used in paints and as an antiseptic in zinc ointment.

ZINC SULFATE (WHITE VITRIOL): A colourless crystalline water soluble powder with the chemical formula ZnSO_4 . It occurs naturally as the mineral goslarite and can be prepared by reacting zinc with sulfuric acid. It is made by heating zinc sulfide ore in air and dissolving out and recrystallising the sulfate. The common form is the heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$). This loses water above 30 °C (303 K) to give the hexahydrate and more water is lost above 70 °C (343 K) to form the monohydrate. Zinc sulfate is used medicinally as an emetic and astringent, as a fungicide and in wood and skin preservatives.

ZINC SULFIDE: A fluorescent white to yellowish inorganic compound, with the chemical formula ZnS . It is the main form of zinc and usually is black because of various impurities. It is found in nature, where it mainly occurs as the mineral sphalerite. It can be produced by igniting

a mixture of zinc and sulfur. It is insoluble in water and can also be produced in a precipitation reaction. It is used especially as a white pigment and a phosphor.

ZINC-CARBON CELL: A type of primary cell made from a carbon and zinc electrode in a paste of zinc chloride and ammonium chloride electrolyte. Manganese dioxide is included in the electrolyte as a depolariser that stops the formation of gas bubbles. It is packaged in a zinc can that serves as both a container and negative terminal. It is used in power devices such as remote controls, flashlights, clocks or transistor radios.

ZINCATE: An alkali solution prepared from dissolving zinc metal, zinc hydroxide or zinc oxide which contains various anionic species such as $\text{Zn}(\text{OH})_4^{2-}$. Such solutions are used in the plating industry. 2. A salt containing $\text{Zn}(\text{OH})_4^{2-}$, also called the **tetrahydroxozincate ion**. Examples include calcium zincate $\text{CaZn}(\text{OH})_4 \cdot 2\text{H}_2\text{O}$ or $\text{Na}_2\text{Zn}(\text{OH})_4$ or the polymeric anion $[\text{Zn}(\text{OH})_3]^-$ in, for example, $\text{NaZn}(\text{OH})_3 \cdot \text{H}_2\text{O}$. 3. An oxide containing zinc and a less electronegative element, for example, Na_2ZnO_2 .

ZINCITE: Deep red-orange mineral, mainly zinc oxide, which is an important zinc ore. Its hardness is 4 and its relative density is 5.6. Zincite is rare except at the zinc deposits at Franklin and Sterling Hill Mines in New Jersey, in the United States, associated with franklinite and willemite. Synthetic zincite crystals are available as a by-product of zinc smelting.

ZINETAC (HYDROCHLORIDE; RANITIDINE; ZANTAC): The name for an anti-ulcer drug with the chemical formula $\text{C}_{13}\text{H}_{22}\text{N}_4\text{O}_3\text{S}$. It is a histamine H_2 -receptor antagonist that inhibits stomach acid production. It is used to treat peptic ulcers and relieve heart burns.

ZINO: One of the fermion partners of existing bosons including gluino, photino and wino. The boson partners of existing fermions have names like selectron, squark and slepton. In the supersymmetry theories, every boson has a corresponding fermion partner and every fermion has a corresponding boson partner.

ZIP DISK: High capacity floppy disk with capacity of several hundreds megabytes (MB).

ZIP FILE: A file that is compressed for transfer to another computer. Zipped files normally have .zip at the end of the file name to indicate that it is zipped. It takes up less hard drive space and takes less time to send. To use it, you need to unzip it, using special programs such as "WinZip" for Windows.

ZIPF'S LAW: A law in statistics, which states that in a population consisting of many different types, the proportion belonging to the most common type is approximately proportional to $\frac{1}{n}$. It was put forward by George Kingsley Zipf (1920-1950), a Harvard University linguistics professor in the context of the frequencies with which words appear in texts.

ZIRCON: A naturally occurring zirconium silicate compound with the idealised composition ZrSiO_4 and one of the chief sources of the element zirconium. It has a hardness of 7.5 and a specific gravity of 4.2 to 4.86 on the Moh's scale. It may be red, brown, yellow or green, depending on the quantity of other materials present in the silicate. It resists weathering and accumulates in residual deposits and in beach and river sands. The transparent colourless or tinted varieties are popular as gemstones.

ZIRCONIUM: A grey-white metallic transition element in Group 4 (IVB) of the Periodic Table with the symbol Zr, atomic number 40 and atomic weight 91.22, melts at about 1855 °C (about 2128 K), boils at about 4409 °C (about 4682 K) and has a relative density of 6.5. The metal burns in air at 500°C (773 K). Its chief ore is zircon. Zirconium

ranks about 18th in abundance among the elements in Earth's crust. It occurs as crystals and flammable as powder. It is insoluble in water but soluble in hot concentrated acids. It has five natural and six known radioactive isotopes. Extraction is by chlorination which is purified by solvent extraction and reduced with magnesium through the Kroll process. It is used in nuclear reactors and in certain alloys.

ZMODEM: In telecommunications, it is an error detecting FTP (File Transfer Protocol) protocol for modems that provides faster data transfer rates and error detection than Xmodem across the Internet or other communications link. It sends data in 512-byte blocks, compared to Xmodem's 128-byte blocks.

ZODIAC: 1. A sequence of events that repeats itself cyclically. 2. A narrow band in the sky through which the movements of the major planets, Sun and Moon take place. It is astrologically divided into twelve sections by the Babylonians, about 2500 years ago and named after the major constellations such as Aries, the Ram, Taurus, the Bull, Gemini, the Twins, Cancer, the Crab, Leo and the Lion. 3. A chart linking twelve constellations to twelve divisions of the year, used as the astrologer's main tool for analysing character and predicting the future.

ZODIACAL LIGHT: A faint luminous glow in the sky that can be observed at low to middle geographic latitudes on a moonless night on the western horizon after sunset or on the eastern horizon before sunrise. In the Northern Hemisphere it can be observed on spring evenings and autumn mornings. It is caused by the scattering of sunlight absorbed by dust particles in interplanetary space.

ZOIDOGAMY: A type of plant reproduction in which male gametes swim in a film of water to the female gametes. It is found in algae, bryophytes, pteridophytes and some gymnosperms.

ZOMBIE: 1. A term used to describe a process that is doing nothing but utilising system resources. 2. A zombie computer is often a computer or server that has been compromised by a Trojan to help malicious attackers to perform activities such as deleting or modifying files, installing other software, uninstalling software or sending sensitive password and login information to a remote attacker.

ZONA PELLUCIDA: A jelly coat surrounding the primary oocyte made up of a layer of glycoprotein in the mammalian egg cell. It develops into a female reproductive cell ovum.

ZONAL SOIL: Soil that has been under development for a very long period of time and has undergone the maximum effect of climate and vegetation. These soils have mature and well-developed profiles determined mainly by the local climate and vegetation. They have clear horizons and generally cover a wide geographic region or zone.

ZONATE: In botany, marked or coloured in circular rings or zones.

ZONATION: The distribution of the different species of a community into biogeographic zones, which are created by variations in the environment.

ZONE OF ACCUMULATION: Upper level of a glacier where more snow falls than melts.

ZONE OF AERATION: Underground region located between the Earth's surface and water table in which the pore spaces are filled mostly with air with some amount of water. This zone mainly comprises porous soil and rock. The air and water are able to interact with each other, leading to aeration of water.

ZONE OF DIFFERENTIATION: Region in plant stems and roots where cells produced by the apical meristem develop into different cell types of the mature organ.

ZONE OF ELONGATION: Area in plant stems and roots where cells recently produced by the apical meristem grow and enlarge prior to differentiation.

ZONE OF INTOLERANCE: The area outside the geographic range, where a population is absent. In the zone of intolerance the range of conditions such as water, temperature, atmosphere, gravity and nutrients, which an organism needs to survive are absent.

ZONE OF PHYSIOLOGICAL STRESS: The area in a population's geographic range, where members of the population are rare due to physical and biological limiting factors.

ZONE OF SATURATION: A layer of permeable rock in which the cracks and pores are completely filled with water under hydrostatic pressure. As the rainwater keeps penetrating the soil under the influence of gravity through the pores of the aeration layer, it finally reaches the impermeable layer deep below the Earth's surface. The composition of this layer is such that further penetration is restricted and water is effectively trapped inside the pores of the rocks present there.

ZONE REFINING (ZONE PURIFICATION): A technique in which a narrow molten zone is moved slowly along the complete length of the specimen to bring about impurity segregation. It is used to reduce the level of impurities in certain metals, alloys, semiconductors and other materials.

ZONED RESERVE SYSTEMS: Habitat areas that are protected from human alteration and surrounded by lands that are used and more extensively altered by human activity.

ZOO: A place where animals are confined within enclosures or a park where live wild animals from different parts of the world are kept for exhibition and where they are bred and studied by scientists.

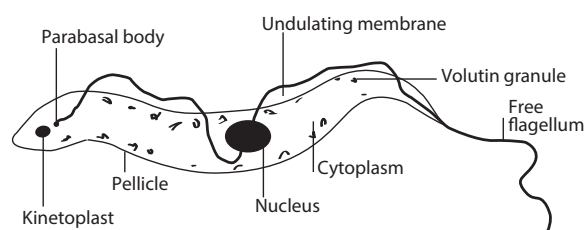
ZOOGEOGRAPHY: The study of the geographical distribution of animals, which is a subdivision of the science of biogeography that is concerned with the detailed description of the distribution of animals and how their past distribution has produced present-day patterns.

ZOOID: 1. Any member of a colony of invertebrates, especially members of the Bryozoa. It may reproduce non-sexually by budding or fission. The term refers especially to one that lives in a colony in which each member is joined to others by living materials such as a coral. 2. An organic cell or organised body that has independent movement within a living organism, especially a motile gamete such as a spermatozoon.

ZOOLOGY: The branch of biology devoted to the study of the animal kingdom (Animalia). The study includes the anatomy, physiology, biochemistry, genetics, parasitology, ecology, evolution, taxonomy and behaviour of animals and several other subdivisions.

ZOOM LENS: A photographic system of lenses in which two or more parts are moved with respect to each other to obtain a continuously variable focal length and hence magnification and allows speedy transition from long shots to close ups.

ZOOMASTIGOTA (ZOOMASTIGINA): One of the major phyla of the kingdom Protista, comprising of parasitic or free living flagellated protozoans. They use their flagella for locomotion. Trypanosoma gambiense-the parasite that causes sleeping sickness is a typical example.



Trypanosoma (example of Zoomastigina)

ZOONOSSES ORDER: A legislation in the United Kingdom under which the presence of conditions such as salmonellosis and brucellosis, which affect both animals and humans, must be reported to regulatory officials. These orders require bovine spongiform encephalopathy (1988) and all isolates of salmonella (1989) from farm animals and birds, their carcasses, products, feed or surroundings to be reported to an officer of Ministry of Agriculture and Fisheries (MAFF). A related order, The Poultry Breeding Flocks and Hatcheries Order 1993 requires the testing of all breeding flocks. In cases where salmonella infection is confirmed after investigation, the flock must be slaughtered. However, in the case of parent broiler breeder flocks, alternatives are offered which include antibiotic treatment of the parent flock or its progeny.

ZOONOSIS: A disease that is transmitted directly from animal to humans without the involvement of a vector or intermediate host. Examples are tuberculosis from cattle and disorders caused by Salmonella bacteria. Of the 1415 pathogens known to affect humans, 61% are zoonotic. If it is transmitted from humans to animals it is called **reverse zoonosis** or **anthroponosis**.

ZOOPHYLOUS: 1. Pollinated by animals other than insects. 2. Having an affinity for animals.

ZOOPHYTE: Any of various invertebrate animals such as a sea anemone or coral or sponge resembling a plant.

ZOOPLANKTON: A component of the plankton, made up of microscopic animals such as protozoans and small crustaceans, jellyfish, worms and molluscs, together with the eggs and larvae of the many animal species that inhabit the water column of oceans and freshwater bodies. Zooplankton lacks the means to counteract transport currents. Important protozoan groups in the zooplankton are dinoflagellates and foraminifera. Zooplankton inhabits all depths of water bodies and constitutes a major link between primary production and higher trophic levels in aquatic ecosystems.

ZOOSPORANGIUM: A sporangium that produces zoospores.

ZOOSPORE: A motile asexual spore that possesses one or more flagella. Released from a sporangium, zoospores are produced by many algae and zoosporic fungi.

ZOOTECHNOLOGY: The use of modern technological advances in animal breeding to increase quality and production.

ZOOXANTHELLAE: Unicellular, yellow-brown photosynthetic dinoflagellates, which live symbiotically in the gastrodermis of marine invertebrates such as corals, jellyfish, and sea anemones.

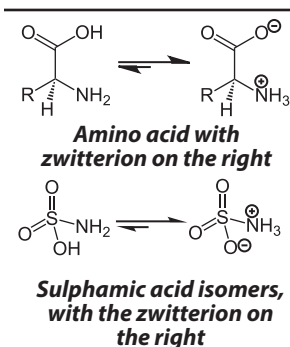
ZORN'S LEMMA (KURATOWSKI-ZORN LEMMA): In mathematics, it is a theorem of set theory that in an ordered set in which every chain has an upper bound, there is an element in the set such that the set contains no element greater than the specified given element. It is named after the German-born American mathematician, Max August Zorn (1906-1993) and Kazimierz Kuratowski (1896-1980), Polish mathematician.

ZOSTEROPHYLLOPHYTES (ZOSTEROPHYLLS): A group of extinct vascular plants that lived during the Devonian period, probably stem-group lycophytes, forming a sister group to the ancestors of the living lycophytes.

ZOUTENDIJK'S METHOD: A prototype feasible direction method of solving constrained optimisation problems by generating feasible descent directions. It is a penalty function concept to yield an efficient and convergent algorithm.

ZWITTERION (AMPHOLYTE): A molecule with a positive and a negative electrical charge at different locations within that molecule.

Examples are amino acid in neutral solution or an ion that has a positive and negative charge on the same group of atoms. Zwitterions can be formed from compounds that contain both acid and basic groups in their molecules. For example, glycine contains both a basic amino group (NH_2) and an acidic carboxyl group (COOH). When these are both ionised in aqueous solution, the acid group loses a proton, while the amino group gains one and the molecule is positively charged at one end and negatively charged at the other.



ZYGNEMAPHYCEAE (CONJUGATOPHYCEAE): A class of the Chlorophyta, whose members are distinguished by their distinctively symmetrical cells, anatomically complex plastids and the lack of free-swimming forms. In addition, they reproduce by conjugation. It contains freshwater filamentous forms, for example, Spirogyra, Zygnema and unicellular forms, the desmids. The desmids are divided into the saccoderm desmids, commonly found in acidic pools and peat bogs; and the placoderm or true desmids, which are a major component of the phytoplankton. The cells of placoderm desmids are characteristically split into two virtually identical halves, joined by a narrow bridge at the centre. The cell wall is highly ornamented and indented.

ZYGODACTYLY: An arrangement of toes found in birds and chameleons, in which two toes face forward and two backward.

ZYGOGENESIS: A reproduction in which male and female gametes and nuclei fuse.

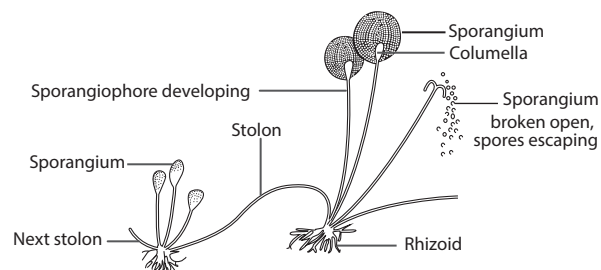
ZYMOGENOUS FLORA: Organisms found in soils in large numbers immediately after the addition of readily decomposable organic materials. Their presence in a given habitat is influenced by the availability of particular nutrients.

ZYGMORPHIC: A structure with a bilateral symmetry, so that a line drawn through the middle of one plane will produce a mirror image on the other side. Examples of zygomorphic flowers are snapdragon orchid and sweet pea. Zygomorphic flowers are also known as **irregular flowers**.

ZYGYMYCETES: A class of terrestrial fungi in the phylum **Zygomycota**, which includes two orders, Mucorales and Entomophthorales, in which resistant spherical spores are formed during sexual reproduction. They are mostly terrestrial in habitat, living in soil or on decaying plant or animal material. Some are parasites of plants, insects and small animals, while others form symbiotic relationships with plants. They produce zygospores. Zygomycetes include bread moulds.

ZYGYMYCOTA: A phylum of saprotrophic or parasitic fungi that includes many important decomposers, mycorrhizal fungi and moulds, such as those that invade breads and other food products. Zygomycota lacks hyphal cell walls except in reproductive structures. They can reproduce by means of zygospores or asexually by

sporangiospores formed within a sporangium. Members of the phylum are called **Zygomycetes** that include two orders, Mucorales and Entomophthorales.



Structure of a Zygomycete

ZYGOPORE: A spore formed from the union of two gametes in some fungi and green protists. It has a thick resistant wall and enters a resting stage before germination. It helps the organism to overcome unfavourable environmental conditions. In fungi, zygospores are termed **chlamydospores** and are formed after the fusion of hyphae of different mating types. In many eukaryotic algae, including many species of the Chlorophyta, zygospores are formed by the fusion of unicellular gametes of different mating types.

ZYGOE (ZYGOCTE): The product formed when a female gamete is fertilised by a male gamete. In other words, it is the cell as a result of the fusion of two haploid nuclei during fertilisation until the first cleavage. In plants with flowers, a zygote results when the male nucleus in the pollen tube fuses with the ovum in the embryo sac. The zygote grows to become a seed. In mammals, a zygote results when a sperm fertilises an ovum. The zygote grows into a young offspring.

ZYGOENE: The second phase of the first prophase of meiosis, in which the two leptotene chromosomes undergo pairing by the formation of synaptonemal complexes to form a bivalent. During this process, intimate contact is made between identical regions of homologues in a process which includes proteins and DNA being organised to form a synaptonemal complex.

ZYMASE: The enzyme present principally in yeasts which induces fermentation of sugars to form ethanol and carbon dioxide. Zymase was first isolated from the yeast cell in 1897 by German chemist and zymologist, Eduard Buchner (1860 - 1917) who fermented sugar in the laboratory without living cells. He was awarded the 1907 Nobel Prize in Chemistry for his work on fermentation.

ZYMOGEN (PROENZYME): An inactive enzyme that, following secretion, is chemically altered to the active form of the enzyme or the inactive precursor of an enzyme, especially one secreted by living cells and activated by an acid, another enzyme or other catalytic means.

ZYMOGENIC CELL (CHIEF CELL; PEPTIC CELL): A cell that lines the lumen of the gastric glands of the stomach and secretes the enzymes pepsin and rennin.

ZYMOGENOUS FLORA: Micro-organisms that respond rapidly by enzyme production and growth when simple organic substrates become available.

SELECTED BIBLIOGRAPHY

- A & C Black Publishers Ltd. (2006).** *Dictionary of Agriculture*. A & C Black Publishers Ltd., London.
- Abbey, T. K., Alhassan, B., Ameyibor, K., Esiah, J. W., Nyavor, C. B., Seddoh, S. K. & Wiredu, W. (2001).** *Ghana Association of Science Teachers Integrated Science for Senior Secondary Schools*. Unimax Macmillan Ltd., Accra.
- Abbot, A. F. (1977).** *Ordinary Level Physics*. Heinemann Educational Publishers, Harlow/United Kingdom.
- Abdelnoor, R. E. Jason and Chandler, S. (2006).** *A Mathematics Dictionary*, Nelson Thornes Ltd., Cheltenham, GL53 7th, UK.
- Ackerman, Dianne (2011):** Play is our brain's favourite way of learning. <https://www.doodlebugs.com.play>
- Addo-Quaye, Albert, I. A. et al (2004).** *General Agriculture for West African Senior School Certificate*. SEDCO Publishing Ltd, Accra
- Aggrey-Fynn, E. (2005).** *Farm Management*. Ghana Library Board Cataloguing in Publication Data, Accra.
- Agyei, D. (2010):** Information Communication Technology use in Mathematics, <http://www.slideshare.net/ddagyei/ictuse-in-the-teaching-of-mathematics>
- Anamuah-Mensah, Jophus, et al (2014):** Inspiring Science and Mathematics Education in Ghana: Report of the Committee that Reviewed Basic Science and Mathematics Education in Ghana, Ministry of Education, Accra, Ghana
- Appiah, D. Boateng (2015):** Gamification in Education: Improving Elementary Mathematics Through Engagement in Hybrid Learning in the Classroom. M.Phil Thesis, Faculty of Art, KNUST.
- Appleton, Ken (2005).** *How Do Beginning Primary School Teachers Cope With Science? Toward An Understanding Of Science Teaching Practice. Research In Science Education Research In Science Education ... Teaching Practice*. March 2003, (vol. 33), Issue 1, pp1-25: 33:1. doi: 10.1023/A:1023666618800.
- Arbonnier, M. (2004).** *Trees, Shrubs and Lianes of West African Dry Zones*. Margraf Publishers GmbH, Paris/France.
- Arkins, P. W. (1940).** *Chemistry* (1st Edition). W.H. Freeman & Co., New York.
- Ayres, F. (1950).** *Theory and Problems of Differential and Integral Calculus*. Schannm Publishing Company, New York.
- Ayres, F. (1954).** *Theory and Problems: Plane and Spherical Trigonometry*. McGraw-Hill Book Company, New York.
- Asiedu, P. & Baah-Yeboah, H.A. (2011).** *Physics for Senior High Schools in West Africa*, Aki-Ola Publications, Accra.
- Awuku, K. A. et al (1991).** *Senior Secondary School Agriculture and Environmental Studies*. Ministry of Education, Accra.
- Backhouse, J. K., Houldsworth, S. P. T. & Cooper B. E. D. (1995).** *Pure Mathematics Book 1* (New Edition). Longman, Hong Kong.
- Borowski E. J. & Borwein J. M. (1989).** *Collins Dictionary of Mathematics*. Harper Collins Publishers, Great Britain.
- Bostock, L. & Chandler, S. (1991).** *Mechanics and Probability*. Stanley Thornes Publishers Ltd, Leckhampton, UK.
- Bostock, L. & Chandler S. (1984).** *Pure Mathematics 2*. Stanley Thornes Publishers Ltd., Old Station Drive, Leckhampton, Cheltenham GL53 ODN, U.K.
- Brady, J. E. & Holum, J. R. (1993).** *Chemistry: The Study Of Matter And Its Changes*. John Wiley and Sons, New York.
- Brithaupt, J. (1990).** *Understanding Physics for Advanced Level*, (2nd Edition). Stanley Thornes Publishers Ltd., Hong Kong.
- Brown, T. L.; LeMay, H. E.; Bursten, B. E. & Burdge, J. J. (2003).** *Dictionary of Chemistry* (2nd Edition). McGraw-Hill Inc, New York.
- Brusca, R. C. & Brusca, G. J. (2003).** *Invertebrates*. Sinaner Association Inc., Massachusetts/United States of America.
- Burton M. D. (2002).** *Elementary Number Theory*. Mc Graw-Hill Companies Inc., New York, NY.
- Butani, D. K. (2006).** *Dictionary of Science*. Academic Publishers, New Delhi, India.
- Campbell, Neil, A. (1991).** *Biology* (2nd Edition), Benjamin/Cumminings Publishers, U.S.A.
- Carey, Francis A. (1996).** *Organic Chemistry* (3rd Edition). The McGraw-Hill Companies Inc., New York.
- Carlson, J. D. et al. (Ed)(1990).** *Electrotheological Fluids: Proceedings of the Second International Conference on ER Fluids*. Wisconsin/U.S.A.
- Chang Raymond (2000).** *Chemistry* (8th Edition), McGraw-Hill Companies Inc., New York.
- Clapham C. & Nicholson, J. (2009).** *Oxford Concise Dictionary of Mathematics*. Oxford University Press Inc., New York.
- Clark, J. O. E. and Hemsley, William (2003).** *Mini Dictionary of Chemistry*. Midway Publishing Ltd, England.
- Clark, J. O. E. and Hemsley, William (2006).** *Mini Dictionary of Biology*. Midway Publishing Ltd., England.
- Clark, J. O. E. and Hemsley, William (2006).** *Mini Dictionary of Mathematics*. Midway Publishing Ltd., England.
- Clark, J. O. E. and Hemsley, William (2006).** *Mini Dictionary of Physics*. Midway Publishing Ltd., England.
- Clarke, S. P. (1974).** *Calculus and Analytic Geometry*. DC Heath and Company, USA.
- Claude, A. V., Eldra, P. S. & Davies, P. W. (1985).** *Biology*. CBS College Publishing, New York/U.S.A.
- Cleveland P.H. Jnr, et al. (2000).** *Animal Diversity* (2nd Edition), McGraw-Hill Companies Inc., U.S.A.
- Cleveland, P. H., Roberts, L. S. & Larson, A. (2001).** *Integrated Principles of Zoology* (11th Edition), McGraw-Hill Companies Inc., U.S.A.
- Coleta, Vincent P. (1983).** *College Physics*. Saunders College Publishing, Philadelphia.
- College Physics**, Opentext College (2013).
- Collins Dictionary of Science (2005).** Harper Collins Publishing, Westerhill Road, Bishopbriggs, Glasgow G64 2QT.
- Columbia Encyclopedia.** *The Columbia Electronic Encyclopedia* (6th Edition). Columbia University Press, U.S.A.
- Cooper, Elmer L. (1990).** *Agriscience Fundamentals and Applications*. Delmar Publishers Inc., Elbany, NY.
- Daintith, John (Ed.), (2005).** *Oxford Dictionary of Physics*. Oxford University Press, Oxford.
- Daintith, J. (2008).** *A Dictionary of Chemistry* (6th Edition). Oxford University Press, Oxford, UK.

- Deeson, E. (1991).** *Physics Basic Facts*. HarperCollins Publishers, Glasgow, Scotland.
- Dei-Ohemeng & Emmanuel Kwabena (2008).** *Integrated Science Dictionary for Schools*. Mena Book Services, Accra.
- Duncan, Tom (1990).** *Physics for Today and Tomorrow* (2nd Edition). Burtler and Tanner Ltd., London.
- Duncan, Tom (2000).** *Advanced Physics* (2nd Edition). Hodder Murray, UK.
- Everitt, B. S. (2006).** *The Cambridge Dictionary of Statistics* (3rd Edition). Cambridge University Press, Cambridge.
- Ewusie, Yanney J. (2002).** *Tropical Biological Drawings* (2nd Edition). Afram Publications, Ghana Ltd.
- Fennema, Owen R. (1996).** *Food Chemistry* (3rd Edition). Marcel Dekker, Inc., NY, U.S.A.
- Frank, Harry & Althoen, Stephen C. (1995).** *Statistics - Concepts and Applications*. Cambridge University Press, UK.
- Gbedemah, R. P. (2008).** *Integrated Science Dictionary and Facts Made Simple for JHS*. My Last Hour Publications, Accra.
- Goodrum, D.; Hackling, M. W. & Rennie, L. J. (2001:11):** *Science Teaching and Learning in Australian Schools: Results of a National Study*. Published by DETYA. link.springer.com
- Green, S.L. (1980).** *Advanced Level Pure Mathematics*. University Tutorial Press Ltd. Slough, UK.
- Gribbin J. (1998).** *An Encyclopedia for Particle Physics*. The Free Press, New York.
- Halliday, D.; Resnick, R. & Walker, J. (2011).** *Fundamentals of Physics* (9th Ed.). New Jersey, John Wiley & Sons.
- Harris, G. J. & Harris, M. W. (1994).** *Plant Identification Terminology*. Spring Lake Publishing Ltd., USA.
- Hecht, Eugene (1994).** *Physics*. Brooks/Cole Publishing Company, USA.
- Hine, Robert & Martin, Elisabeth (Eds)(2005).** *Oxford Dictionary of Biology*. Oxford University Press, Oxford, OX2 6DP.
- Hirsch, E.D.; Joseph F. Kett & James Trefil (Eds) (2005).** *Dictionary of Cultural Literacy: Science* (3rd Edition). Published by Houghton Mifflin, Kansas/U.S.A.
- Hunt, Andrew (2009).** *A-Z Chemistry*. Philip Alan Updates, United Kingdom.
- Iloje, S. O. (2008).** *Senior Secondary Certificate Practical Biology*. Longman Nigeria Plc.
- Indge, Bill (2012).** *A-Z Biology* (4th Edition). Philip Alan Publishers, United Kingdom.
- Isaac, Alan (Ed) (2005).** *Dictionary of Physics*. Magpie Books (Constable and Robinson Ltd), London, UK.
- IUPAC (1976).** *Manual of Symbols and Terminology for Physicochemical Quantities and Units*.
- IUPAC (1996).** *Glossary of Terms used in Photochemistry* (IUPAC Recommendations).
- IUPAC-IFCC (1996).** *Glossary of Terms in Quantities and Units in Clinical Chemistry*.
- Jones, Loretta, L. & Atkins, Peter W. (1997).** *Chemistry- Molecules, Matter and Change* (3rd Edition), W.H. Freeman and Company, New York.
- Kindle H. J. (1950).** *Plane and Solid Analytic Geometry*. (Schaum's outline series).
- Kozloff, E.N. (1990).** *Invertebrates, "ClassTurbellaria"*. Sanders College Publishing, Fortworth.
- Kreyszig E. (1983).** *Advanced Engineering Mathematics*. John Wiley and Sons New York.
- Kriek, J. & Grayson, D. (2009. 185-186).** *A Holistic Professional Development Model For South African Physical Science Teachers*. South African Journal Of Education, 29, 185-203.
- Lackie, J. M. (4th Edition) (2007).** *The Dictionary of Cell and Molecular Biology* (4th Edition). Elsevier Inc. London.
- Lafferty, Peter and Rowe, Julian (1994).** *Dictionary of Science*. Helcon Publishing Ltd., Oxford, UK.
- Landers, R. N. & Callan, R. C. (2011):** Casual social games as serious games: the Psychology of gamification in undergraduate education and employee training. In Ma, M., Oikonomou, A. & Jain L.C (eds.) *Serious Games and Edutainment Applications*. Springer London
- Lartey, Thomas E.; Asiegbor, Isaac K.; Wilson, Genevieve & Asante-Kissiedu, Daniel (2005).** *Basic Education Certificate Examination Agriculture for JSS* (Pupils Book 1, 2 and 3). Sedco Publishing Ltd., Accra.
- Lee, E. Yunker, et al (1986).** *Advanced Mathematical Concepts*. Merrill Publishing Co., Columbus, Ohio, U.S.A.
- Lego Foundation in Support of UNICEF (2018):** Learning Through Play. Unicef.org/sites/default/files/18-12/UNICEF
- Lytle, F. C. and Meyer, R. J. (2005).** *General Zoology* (14th Edition). McGraw-Hill Companies, Inc., U.S.A.
- Mackean, D. G. (1985).** *Introduction to Biology*. John Murray (Publishers) Ltd., London.
- Mader, Sylvia, S. (2006).** *Human Biology* (9th Edition).
- Malik, M.A. (n.d.).** *Electronics*. House of Physics Publications.
- Mangwaya Ezron et. al. (2016).** *Factors Influencing Ordinary Level Students' Academic Performance in Integrated Science: A Case Study of Three Secondary Schools in the Victoria Falls Town of Zimbabwe; International Journal Advances in Social Science and Humanities*. www.ija.ssh.com
- Morrison, R.T. and Boyd.** *Organic Chemistry*. 6th Edition, Prentice-Hall Inc., Englewood Cliffs, N. J., USA.
- Neal, Steve (2007).** *ICT Dictionary for Schools*. EPP Books Services Edition, La, Accra.
- Nelkon, M. & Parker, P. (2005).** *Advanced Level Physics* (5th Edition). Heinemann Educational Books, London.
- Nelkon, M. & Parker, P. (2001):** *Principles of Physics* (7th Edition). S. Chand & Company Ltd., New Delhi.
- Neuinger, H. O. (1996).** *African Ethnobotany Poisons and Drugs*. Chapman and Hall GmbH, Weinheim, Germany.
- New, T. R. (1995).** *An Introduction to Invertebrate Conservation Biology*. Oxford UP.
- Muntean, C. J., (2011):** Raising engagement in E-Learning through gratification. The 6th International Conference on Virtual Learning ICVL2011 (P7) Strada Mihail Cogalniceanu, Nr. I, Cluj-Napoca, Romania
- O'Neil. V.P. (1995).** *Advanced Engineering Mathematics*. Brooks/ Cole Publishing Company, USA.
- Obeng, A. K. B. et al (2003).** *Approacher's Series Integrated Science for Senior High School*. Approachers Ghana Ltd, Kumasi, Ghana.
- Opoku-Agyemang, Jane (2019):** STEM Education in Ghana, starrfm.com.gh

Oredein, Afolakemi & Awodun, Adebisi (2012). *Impact of Teacher Motivational Indices on Science Students' Academic Performance in Nigerian Senior Secondary Schools*. URL: <http://dx.doi.org/10.5539/ies.v6n2p49-50>

Patterson, M.; Frost, G. S.; Gbatei, M. A., & Bloom, S. R (2003). *Pancreatic Polypeptide Reduces Appetite and Food Intake in Humans, Journal of Clinical Endocrinology & Metabolism*. Vol. 88, No. 8, 3989-3992.

Pilbrow, J. R. (1990). *Transition Ion Electron Paramagnetic Resonance*. Clarendon Press, Oxford, England.

Poole, C. P. Jr. (1990). *Electron Spin Resonance: A Comprehensive Treatise on Experimental Techniques* (2nd Edition). New York: Dover.

Rich, Gillian (2002). *Letts Educational Limited*. The Chiswick Centre, 414 Chiswick High Road, London, W4 5TF.

Riley, Peter, D. (2005). *Cambridge Checkpoint Science, Students' Book 1, 2 and 3*. London/United Kingdom.

Roger, G. et al. (1984). *Macmillan Biological Drawings for Tropical Schools*. Macmillan Publishers, Hong Kong.

Rosen, K. N. et al (2002). *Handbook of Discrete and Combinatorial Mathematics*. CRC Press, New York.

Ruppert, E. E. & Barnes R. D. (1994). *Invertebrate Zoology*, (6th Edition). "Class Turbellaria", 211-240, Saunders College Publishing, Fortworth.

Sang, David and Mike Chapple (2009). *A-Z Physics Handbook* (4th Edition). Philip Alan Updates.

Scott, W. A. H. (1991). *Chemistry Basic Facts*. HarperCollins Publishers, Glasgow, Scotland.

Serway, R. A. & Jewett, J. W., Jr. (2010). *Physics for Scientists and Engineers with Modern Physics* (81th Ed.). U.S.A., Brooks Cole.

Silberberg, M. S. (2000). *Chemistry. The Molecular Nature of Matter and Change* (2nd Edition). McGraw-Hill Higher Edu., U.S.A.

Smith, Alan (2003). *Maths. Dictionary for Junior Secondary Schools*. Letts Educational Limited, The Chiswick Centre, 414 Chiswick High Road, London, W4 5TF.

Smith, J.E; Carthy, J.D; Chapman, G; Clark, R.B; Nichols, D. (1971). *The Invertebrate Panorama*. Universe Books, New York.

Twumasi, Deborah Kingsley (2015). *Integrated Science in Scope: Practical Book for Junior High Schools*. Kingawily Publication Ltd, Tema.

Upshall, Michael et al (1993). *Hutchinson Dictionary of Biology*. Helicon Publishing Ltd., London, UK.

Upshall, Michael et al (1993). *Hutchinson Dictionary of Chemistry*. Helicon Publishing Ltd. London, UK.

Upshall, Michael et al (1993). *Hutchinson Dictionary of Physics*. Helicon Publishing Ltd., London, UK.

Upshall, Michael et al (1993). *Hutchinson Dictionary of Science*. Helicon Publishing Ltd., London, UK.

Upton, Graham and Cook, Ian (2008). *Oxford Dictionary of Statistics*. Oxford University Press.

Wertz, J. E. & Bolton, J. R. (1986). *Electron Spin Resonance: Elementary Theory and Practical Applications*. Chapman and Hall, New York.

Yadolah, D. (2008). *The Concise Encyclopedia of Statistics*. Springer, Berlin, Germany.

Yang, Y.S., Wu, M.C. & Wang, H.Y. (2009): Investigating the Determinants and Age and Gender Differences in the Acceptance of Mobile Learning. *British Journal of Educational Technology*, 40 (1,92-118)

Yann Teyssier (2016): E-learning Authoring tools: 5 Advantages of Using Games for Learning. elearningindustry.com/5-advantages-games-for-learning.

Yu-kai Chou (2012): Actionable Gamification. <https://yukaichou.com/gamification>

Zubay, L. G; Parson, W.; William & Van, E. Dennis (1995). *Principles of Biochemistry*, Wm. C. Brown Publishers, USA, Australia, England.

ONLINE RESOURCES

Biology for ID Diploma, Second Edition © C. J. Glegg
2014 Published by Hodder Education.

Biology Glossary: Idaho State Department of Education, Copyright © 2013.

Biology Glossary: A Collection Of Terms From The Science Syllabus Glossaries And Senior Sciences Curriculum Glossary; Queensland Curriculum & Assessment Authority, September 2016.

ENCARTA (MICROSOFT ENCARTA)

Everyday Mathematics Teacher's Reference Manual, By Max Bell, et al., Copyright @ 2012 The McGraw-Hill Companies

Excel Math Glossary: <http://excelmath.com/>

GENERAL AGRICULTURAL SCIENCE TERMINOLOGY: MISSISSIPPI SOYBEAN PROMOTION BOARD (<https://www.mssoy.org/>).

GLAST SCIENCE GLOSSARY: <http://glast.sonoma.edu/science/gru/glossary.html>.

Glossary of Biology Terms and Definitions, Buzzle – Zoology Terms, <https://biologywise.com/biology-glossary-of-terms-definitions#ZoologyGlossary>.

GLOSSARY OF CLIMATE CHANGE TERMINOLOGY. http://www.seecimateforum.org/upload/document/glossary_on_climate_changes_terminology.pdf

GLOSSARY OF TERMS IN SOIL SCIENCE, RESEARCH BRANCH, CANADA DEPARTMENT OF AGRICULTURE, PUBLICATION: Agriculture Terms & Definitions: https://agclass.nal.usda.gov/glossary_az.shtml.

GLOSSARY OF TERMS IN SOIL SCIENCE, RESEARCH BRANCH, CANADA DEPARTMENT OF AGRICULTURE, PUBLICATION: Agriculture Terms & Definitions: https://ag-class.nal.usda.gov/glossary_az.shtml.

Glossary of the Common Agricultural Policy (update of 2015), European Union.

Grade 10 Physical Science by Siyavula and volunteers.

IGCSE Biology © Cambridge University Press 2010, <http://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-biology-0610/>

K-5 Definitions of Math Terms: <http://www.vestal.stier.org/Downloads/K-5glossary2.pdf>

Kansas Science Education Standards Glossary, (www.ksde.org).

Mathematics Glossary: Dynamic Learning Maps (DLM).

Physical Science GLOSSARY, <http://www.stmargaretsch.org/resources/Physical-Science.pdf>

PHYSICS TERMS, DEFINITIONS & UNITS: WELSH JOINT EDUCATION COMMITTEE

Required Vocabulary – IB 12 Higher Level, Physics and Physical Measurement, <https://www.coursehero.com/file/16185147/vocabulary/>

Science glossary 6th grade

https://www.alanpedia.com/science_glossary_6th_grade/science_glossary_6th_grade.html

Science glossary grade 7 & 8

OTHER ONLINE RESOURCES

<http://www.aaamath.com>

<http://allchemicals.info/index/actio>

<http://bbc.co.uk/weather/weatherwise>

<http://bt.cdc.gov/radiation/glossary.asp>

<http://chemguide.co.uk>

<http://chemicool.com>

<http://chemistry-dictionary.com/defi>

<http://chemistrydictionary.com>

<http://chemwiki.ucdavis.edu>

<http://collinsdictionary.com>

<http://davidddarling.info/encyclopedia/alphindexh.html>

<http://diracdelta.co.uk/science/sour>

<http://electropedia.org/iev/iev.nsf/>

<http://en.m.wikipedia.org>

<http://encyclo.co.uk/local/21001>

<http://encyclo.co.uk/search.php>

<http://expertglossary.com/definition/>

<http://googlesearch.com>

<http://instron.co.uk/wa/resourcecent>

http://ionactive.co.uk/glossary_atoz

<http://khanacademy.org>

<http://lectlaw.com/def/d044.htm>

<http://medilexicon.com/medicaldictionary>

<http://medterms.com/script/main/art>

<http://memidex.com>

<http://merriam-webster.com>

<http://mtwilson.edu/ao/>

<http://physicalgeography.net/physgeo>

<http://physics.about.com/>

<http://probertencyclopaedia.com/brow>

<http://scienceworld.wolfram.com/physics/Fall.html>

<http://skyview.gsfc.nasa.gov/help/dictio>

<https://simplicable.com/new/complex-systems>

<http://thefreedictionary.com>

<http://www.themathleague.com/>

<tps://www.thoughtco.com/criteria-for-science-and-scientific-theories-250570>

<https://www.timelinetrust.com/sat.html>

<http://tulane.edu>

<http://users.zetnet.co.uk/computing/>

<http://webdictionary.co.uk/definitionary.co.uk/>

<http://Wikipedia Free Encyclopedia>

<http://WolframMathworld.com>

<http://wordinfo.info/unit/550/>

<http://wordnetweb.princeton.edu/perl/web>

<http://www.en.wikipedia.org/wiki>

<http://www.encyclo.co.uk/visitor-contrib>

www.amathsdictionaryforkids.com

www.mathwords.com

<https://www.osmosis.org/plans>

<https://www.supportatosmosis.org>

<https://www.osmosis.org/library>

<https://www.osmosis.org/love>

<https://words.usask.ca/gmcte/author/can/793>

[blogspot.com/2011/07/abl](https://www.blogspot.com/2011/07/abl)

<https://osmosisgames.org/>

www.worthofread.com/

<https://targetstudy.com/>

<https://www.quora.com>.

<https://www.quora.com>> is abacus useful...

APPENDIX 1A

CARDS (NUMBERS) FOR PLAYING THE OSMOSIS GAME OF "FUNTIME WITH SCIENCE"
(DETACH/CUT EACH NUMBER OUT FOR GAME)

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32

APPENDIX 1B

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

THE RAPID
EFFECTIVE
LEARNING
PROCESS

APPENDIX 1C

CARDS (NUMBERS) FOR PLAYING THE OSMOSIS GAME OF "FUN TIME WITH SCIENCE"
(DETACH/CUT EACH NUMBER OUT FOR GAME)

				
33	34	35	36	
37	38	39	40	
41	42	43	44	
45	46	47	48	
49	50	51	52	
53	54	55	56	
57	58	59	60	
61	62			

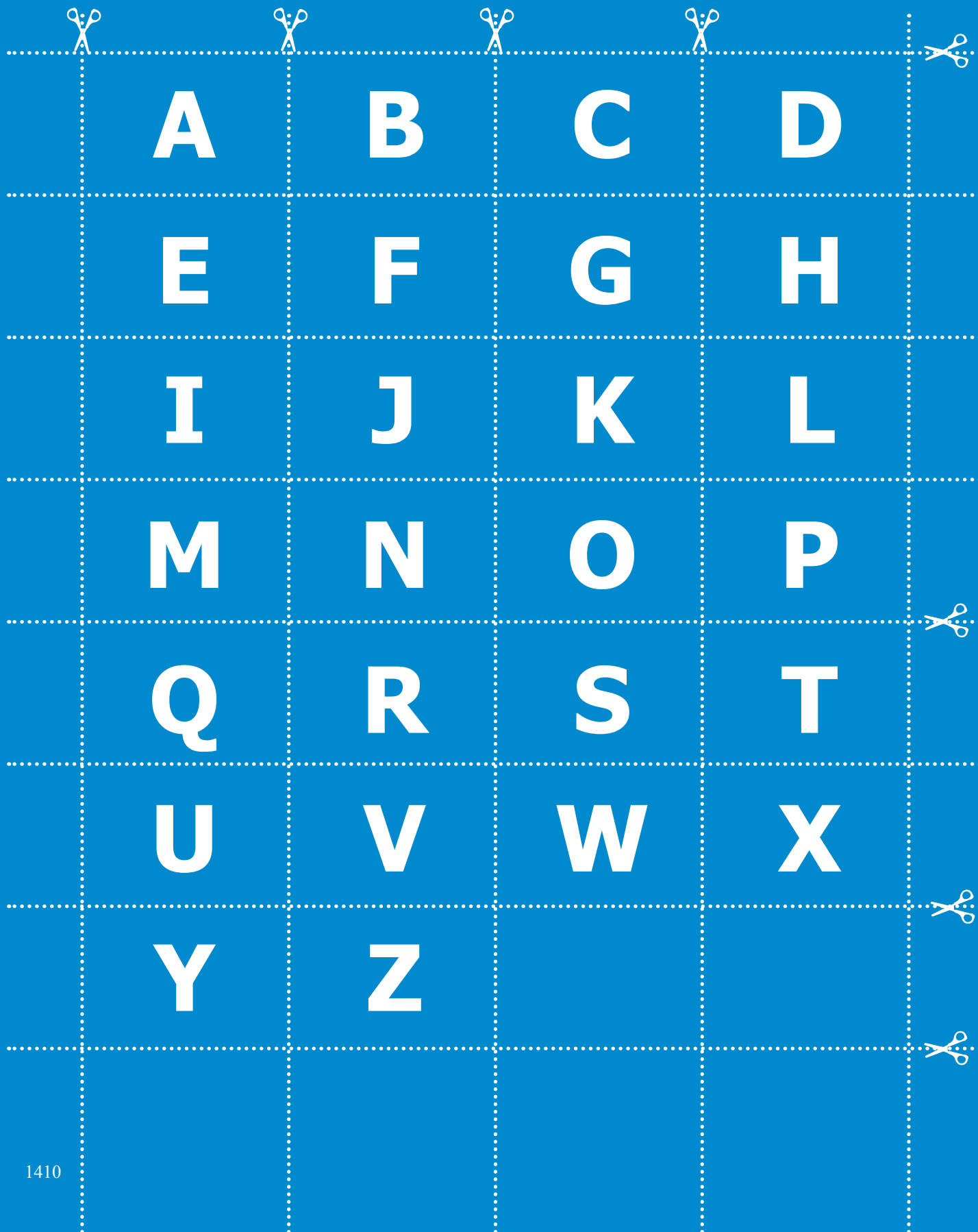
APPENDIX 1D

THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS

APPENDIX 2A

CARDS (LETTERS OF THE ALPHABET) FOR PLAYING THE OSMOSIS GAME OF
"FUN TIME WITH SCIENCE"

(DETACH/CUT EACH LETTER OUT FOR GAME)



APPENDIX 2B

THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS
THE RAPID EFFECTIVE LEARNING PROCESS	THE RAPID EFFECTIVE LEARNING PROCESS		

Page 10 of 10

SCORING CHART FOR PLAYING THE OSMOSIS GAME OF "FUN TIME WITH SCIENCE"

[illegible]

SCORING CHART FOR PLAYING THE OSMOSIS GAME OF “FUN TIME WITH SCIENCE” (CONTINUED)

No of Games Played	Dictionary Column for word search	Dictionary Skills Score	WORDS FOUND	NAME OF PLAYER 1:							NAME OF PLAYER n:						
				Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score	Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score
14 TH	N																
15 TH	O																
16 TH	P																
17 TH	Q																
18 TH	R																
19 TH	S																
20 TH	T																
21 ST	U																
22 ND	V																
23 RD	W																
24 TH	X																
25 TH	Y																
26 TH	Z																
		TOTAL SCORES															

SCORING CHART FOR PLAYING THE OSMOSIS GAME OF “FUN TIME WITH SCIENCE”
CONTINUED

ADVERT SPACE

Contact

+233 (0)24 891 9104